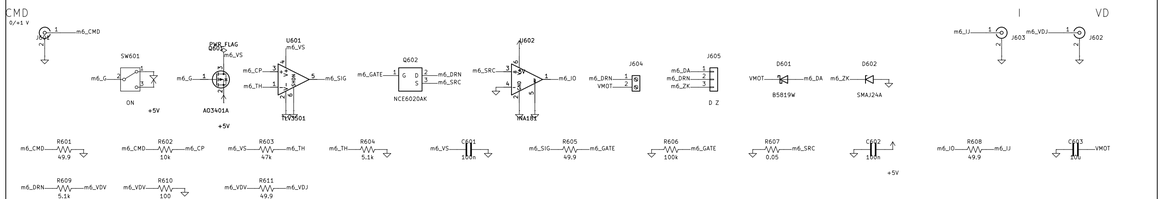
m6 solenoid, NCE6020AK

tutorial Motors (+5 V, VMOT)

cells (0.0), (1.0)

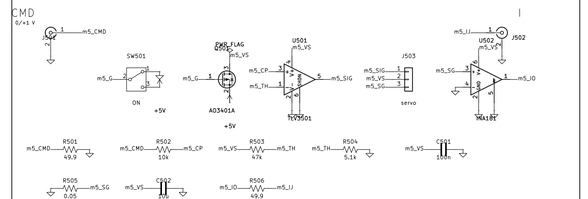


EXPECT: the coil on while CMD is high: I = its current, 1 V/A (up to 2 A); VD = the drain / 52: the flyback's spike, by the jumper
CMD (0 / +1 V) through 10 k into a TLV3501 drives an NCE6020AK (60 V) that switches the coil to ground; VMOT on the coil.
Jumper 1–2: a Schottky to VMOT (slow decay); 2–3: a 24 V TVS (fast decay); in-between: FET avalanches. I: 1 V/A, VD drain / 52.

m5 servo

tutorial Motors (+5 V)

cell (2.0)

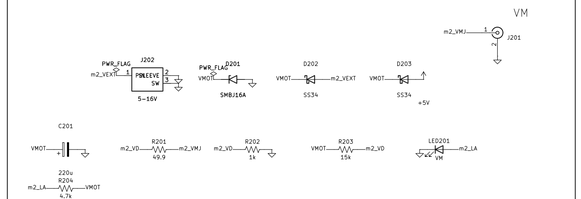


EXPECT: a servo at the angle a 1–2 ms pulse every 20 ms sets (CMD 0 / +1 V): I = the servo's current, 1 V/A
CMD (0 / +1 V from a DAC) through 10 k into a TLV3501 (threshold 0.49 V) makes the 5 V pulse a servo wants; the servo's ground returns through 0.05 ohm and an INA181A1 (x20) puts 1 V/A on I. Slew the angle and watch the current.

m2 motor supply, 5–16 V in

tutorial Motors

cell (3.0)

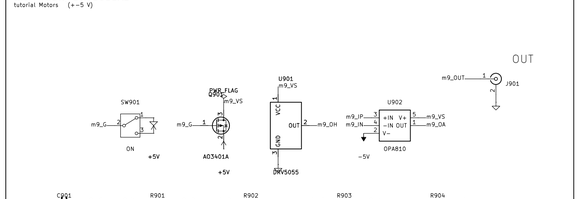


EXPECT: VMOT = the larger of the barrel jack or terminal (5–16 V) and SUB 5 V, each less a Schottky's 0.3–0.5 V: VM jack VMOT / 16
Barrel jack (centre +) or terminal, 5–16 V: each through an SS34 into VMOT, as is USB's 5 V, so VMOT is about 4.6 V with no supply. SMBJ16A and 220 uF on VMOT take the braking energy a motor returns. VM: VMOT / 16 at high impedance.

m9 hall, DRV5055A3

tutorial Motors (+−5 V)

cell (0.1)

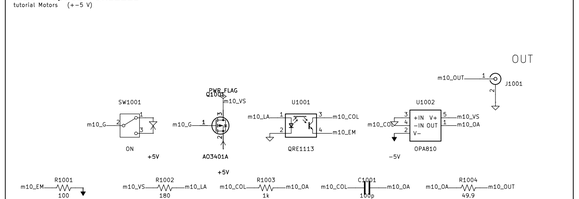


EXPECT: OUT = 9.75 mV per mT at the sensor (+−0.83 V at +−85 mT), 0 V with no field: about 20 kHz wide
DRV5055A3: 25 mV/mT about half its 5 V supply, ratiometric. The OPA810 subtracts the same half (20 k / 20 k) and scales by 0.39: OUT 9.75 mV/mT. A magnet's distance, a magpie's gap, a gear tooth: the sensor is the small part top left.

m10 optical, QRE1113

tutorial Motors (+−5 V)

cell (1.1)

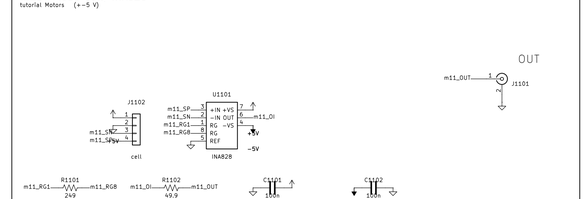


EXPECT: OUT = 1 k x the phototransistor's current: about 0.1–0.9 V with white paper 1 mm above the sensor, near 0 V with nothing
QRE1113GR: an IR LED (21 mA through 180 ohm) and a phototransistor side by side. The OPA810 holds the collector at 0 V (emitter on −5 V through 100 ohm): OUT = 1 k x Ic. Distance in the first millimetres, a mark on a wheel, a line.

m11 load cell, INA828

tutorial Motors (+−5 V)

cell (2.1)

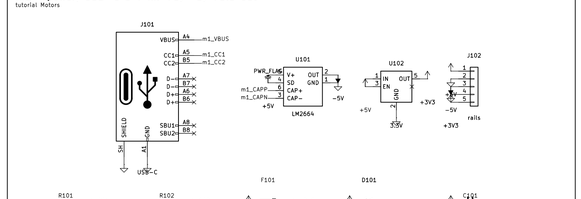


EXPECT: OUT = 201 x the bridge's output: about 1 V at full load for a 1 mV/V cell; about 10 kHz wide
INA828 at G = 1 + 50 k / 249 = 201 on +−5 V, REF on ground: the bridge excited from +5 V. Weigh, feel a force, close a force loop with the servo or the linear amplifier. Cable: 1 red E+, 2 black E−, 3 white S−, 4 green S+.

m1 power, USB-C 5 V in: +5, −5, +3.3 out

tutorial Motors

cell (3.1)

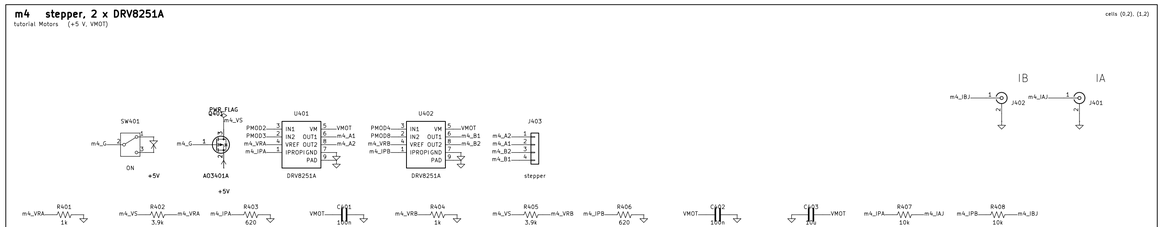


EXPECT: +5 V from the USB-C (up to 3.5 A), −5 V at 40 mA, +3.3 V at 300 mA; the green LED says 5 V is in, the red one that −5 V is up
USB-C power only: 5.1 k on each CC asks any source for 5 V. A 2 A PTC, a 5 V TVS (clamps at 9.2 V) and 20 uF.
The LM2664 charge pump makes −5 V for the op-amp rails (40 mA; four tiles at 10 mA): it is not switched yet, an op amp with its −5 V removed draws nothing. The TPS7A2033 makes 3.3 V for the comparator and the lines that meet the FPGA.
The socket along the top carries +5, ground, −5, ground, +3.3: the bench's rails for a breadboard experiment.

m4 stepper, 2 x DRV8251A

tutorial Motors (+5 V, VMOT)

cells (0.2), (1.2)

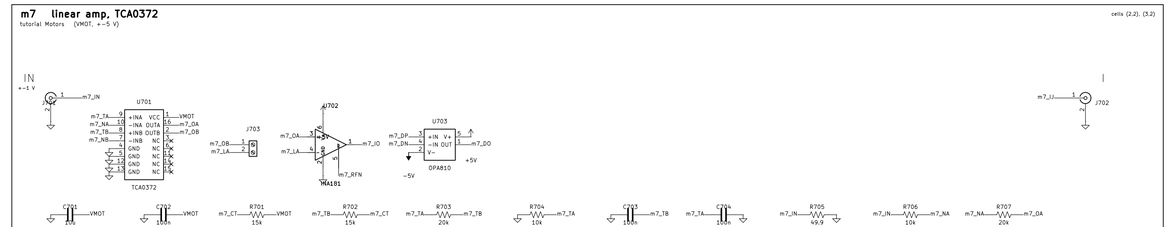


EXPECT: full, half or PWM microsteps at the rate the FPGA sets (IN1, IN2 per coil): IA, IB = the coil currents' size, 0.98 V/A, 1 A limit
Two DRV8251A from VMOT: coil A IN1 on IO2, IN2 on IO3; coil B IN1 on IO4, IN2 on IO8 (both low: coast, then sleep).
VREF 1.0 V and 620 ohm on IPROPI: 1 A limit, 0.98 V/A through 10 k onto IA and IB. Stepper on the XH: A+, A−, B+, B−.

m7 linear amp, TCA0372

tutorial Motors (VMOT, +−5 V)

cells (2.2), (3.2)

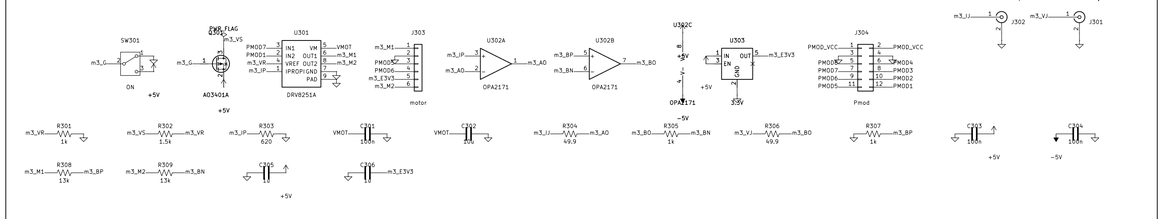


EXPECT: the load at 4 x IN (+−4 V for +−1 V), about VMOT / 2 at each end: I = the load's current, 1 V/A (+−1 A)
A TCA0372 bridge from VMOT: side A = VMOT/2 − 2 IN, side B = VMOT/2 + 2 IN, so the load sees 4 x IN. Up to 1 A and about 1 W in the chip: 8 ohm at 5–6 V, 30 ohm or more at 12 V. 0.1 ohm + INA181 + OPA810: I = 1 V/A. A voice coil, a maglev magnet.

m3 DC motor, DRV8251A

tutorial Motors (+−5 V, VMOT)

cells (0.3), (1.3)

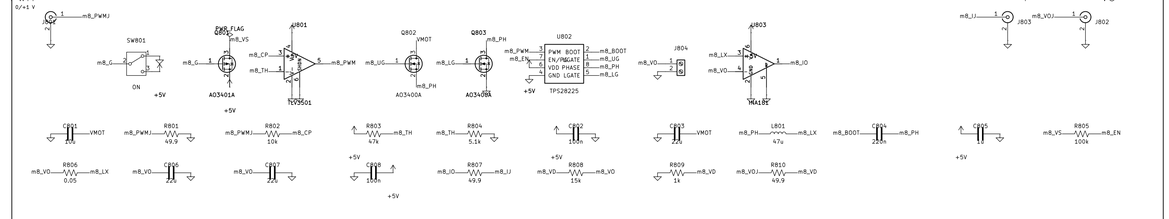


EXPECT: the motor at the duty and direction IO7 (IN1) and IO1 (IN2) set: I = its current, 0.98 V/A; V = its terminal voltage / 13 (back-EMF in coast)
DRV8251A from VMOT: IN1 (IO7) and IN2 (IO1) choose forward, reverse, brake (both high) or coast (both low: then sleep).
IPROPI (620 ohm, 0.98 V/A, 2 A limit) buffered onto I: (M1 − M2) / 13 onto V. Encoder A on IO5, B on IO6, its own 3.3 V.

m8 buck stage, TPS28225

tutorial Motors (+5 V, VMOT)

cells (2.3), (3.3)



EXPECT: VO = the duty cycle x VMOT at the PWM's frequency (100–500 kHz); I = the inductor's current, 1 V/A (0 when it dips below zero): VO jack VO / 16
PWM (0 / +1 V) squared by a TLV3501 into a TPS28225 driving two AO3400A from VMOT: 47 uH into 2 x 22 uF. Close the loop in the FPGA from VO (VO / 16) and I (1 V/A): a load on the terminal, up to 1 A. The switch holds the driver's EN low.

GND GND
no mounting holes: four stick-on rubber feet, marked on the back

PWR_FLAG
ground loops for the scope's clips, one per channel, in the board's margins

Drawn by scripts/gen_sch.py from boards/amod_m.py: the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
a DC motor with its encoder, a stepper, a servo, a solenoid, a linear amplifier, a buck stage, a hall, a
Board: Jason Gallicchio and Claude | open hardware, CERN-OHL-P-2.0 | github.com/gallicchio/

Sheet: /		
File: amod_m.kicad_sch		
Title: AMOD−M: motors and actuators		
Size: A1	Date: 2026-10-25	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/1